

Experiment D: Permittivity Constant (ϵ) of Dielectric materials

1.1 Objectives

- (i) To observe the relation between charge Q and potential V in a plate capacitor.
- (ii) To determine permittivity of air ϵ_0 .
- (iii) To observe relation between charge Q and the inverse of the distance between the two parallel plates $1/d$ for a constant potential V .
- (iv) To determine the relative permittivity ϵ_r of a solid dielectric medium.

1.2 Brief theory

The induced electric field $\mathbf{E}$ in vacuum (and with a good degree of approximation in air) in a closed Gaussian surface s is given as,

$$\oint_s \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0} \quad (1)$$

where, Q is the charge enclosed by the closed surface s and ϵ_0 is the permittivity of the vacuum.

When the two conducting plates of a parallel plate capacitor are charged by a voltage source, the charges of opposite polarity are drawn towards the surface of the capacitor plates resulting in an induced electric field between the two plates. If the distance between the two plates are sufficiently small compared to the plate surface area, the field lines of $\mathbf{E}$ are always perpendicular to the surface. Fig.1 shows the electric field in a plate capacitor with small distance between the plates as compared to the plate dimensions.

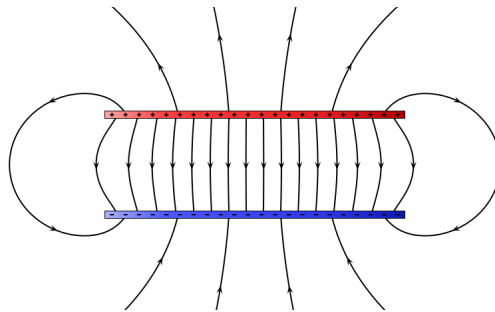


Fig.1: Electric field of a plate capacitor

If the surface area of the plate is A , the magnitude of the electric field found from (1) is

$$E = \frac{Q}{\epsilon_0 A} \rightarrow Q = \epsilon_0 A E \quad (2)$$

The electric potential V between the two parallel plates, can be found from (3).

$$V_{12} = -\int_1^2 \vec{E} \cdot d\vec{l} \quad (3)$$

Considering plate 1 at ground and distance between the two plates d , the electric potential V is found from (3) as,

$$V = -\int_0^d \vec{E} \cdot d\vec{l} = Ed \quad (4)$$

The charge Q of the capacitor is proportional to the voltage V and the proportionality constant C is called the capacitance of the capacitor,

$$Q = CV \rightarrow C = \frac{Q}{V} \quad (5)$$

Replacing Q and V from (2) and (4),

$$C = \epsilon_0 \frac{A}{d} \quad (6)$$

Note from (6) that C is inversely proportional to d . Thus, in a parallel plate capacitor of fixed area A , capacitance C changes if distance d is varied.

Thus,

$$Q = CV = \epsilon_0 \frac{A}{d} V \quad (7)$$

For a constant voltage, inverse distance between the plates $1/d$ and the capacitance C are the measure of the amount of charge Q in a capacitor.

On the other hand, by measuring V , Q , d and A of a capacitor, the permittivity constant ϵ_0 can be found as,

$$\epsilon_0 = \frac{d}{A} \frac{Q}{V} \quad (8)$$

The equation (6), (7) and (8) are valid only approximately due the assumption that field lines are parallel. If the distance d is too large and becomes comparable to plates' dimensions, electric field lines will not be homogenous, and the permittivity found from (8) will divert by large amount from the exact value of $\epsilon_0 = 8.8542 \times 10^{-12} \text{ F/m}$.

If a dielectric material is inserted between the two charged plates, the polarization (formation of dipoles) will occur in the material. The total dipole moment per unit volume is defined as the polarization vector $\mathbf{P}$, which is directly proportional to the applied electric field $\mathbf{E}$, if the dielectric medium is linear and isotropic.

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} \quad (9)$$

Where, χ_e is a dimensionless constant called electric susceptibility.

The effect of polarization of the dielectric is equivalent of induction of a bound surface charge density at the surface of the dielectric material. Consequently, the electric field $\mathbf{E}$ induced by the actual charge Q weakens within the dielectric material.

From the first postulate of electrostatic, the relation of the polarization vector $\mathbf{P}$, with the applied electric field $\mathbf{E}$ and electric field density $\mathbf{D}$ can be found as,

$$\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E} \quad (10)$$

Substituting $\mathbf{P}$ from (9) in (10), it can be shown that,

$$\mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon_0 \epsilon_r \mathbf{E} \quad (11)$$

Where, $\epsilon_r = 1 + \chi_e$ is called relative permittivity of the dielectric media.

If the charge Q in the capacitor is unchanged, the potential between the plates V reduces by the ϵ_r when a dielectric is inserted. If V_{vac} is the potential when the medium between the plates is vacuum (or air), the potential with the dielectric is

$$V = V_{vac} / \epsilon_r \quad (12)$$

Similarly, capacitance $C = \epsilon_r C_{vac}$

Therefore, general form of (7) can be written as,

$$Q = CV = \epsilon_0 \epsilon_r \frac{A}{d} V \quad (13)$$

If the charge obtained with and without dielectric are known, the relative permittivity ϵ_r can be found as

$$\frac{Q_{dielectric}}{Q_{vacuum}} = \epsilon_r \quad (14)$$

Equipment Required

1. Parallel plate capacitor
2. Plastic plate $283\text{mm} \times 283\text{mm}$
3. $10\text{ M}\Omega$ resistor
4. Amplifier
5. High voltage supply
6. 220nF capacitor
7. Voltmeter $0.3\text{-}300\text{V DC}$
8. Connectors and cables

1.3 Procedure

General Description:

The experimental set-up is shown in Fig. 2(a) and (b) and the corresponding wiring diagram is shown in Figure 3.

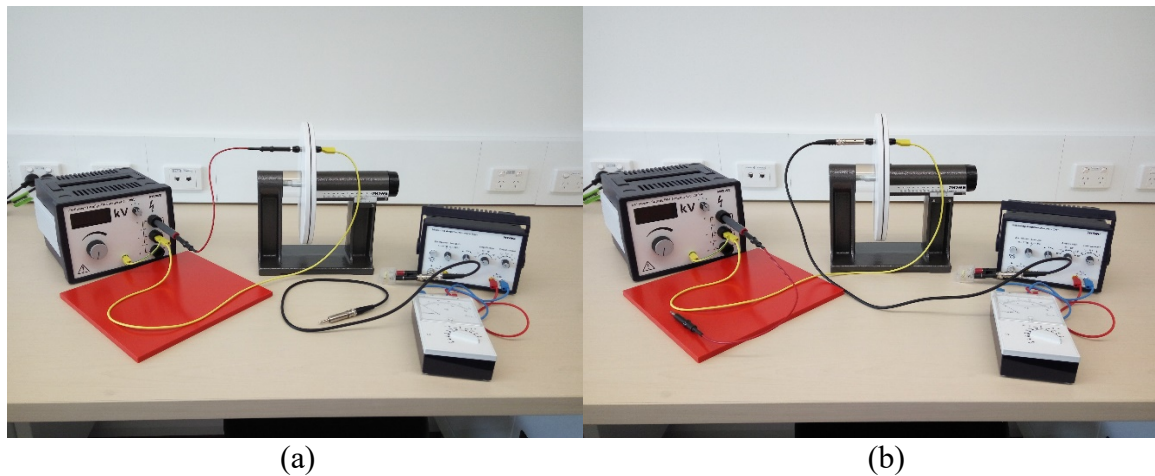


Fig.2 (a) Charging of the plate capacitor and (b) measurement of voltage (charge).

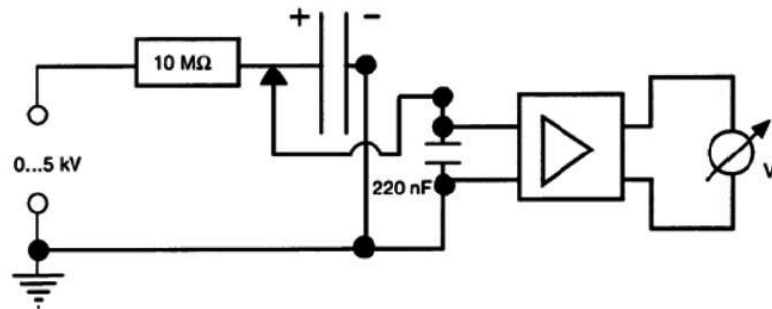


Fig.3 Wiring diagram

The highly insulated capacitor plate is connected to the upper connector of the high voltage supply via a $10\text{M}\Omega$ resistor. Both the middle connector of the high voltage supply and the opposite plate are grounded over the 220 nF capacitor as shown in Fig.2(a). The supply voltage

can be varied by adjusting the turning switch on the unit. The charge on the parallel plate capacitor is measured according to equation (7) using an amplifier. The amplifier is set to high input resistance mode (i.e. Electrometer mode, $R_i \geq 10^{13}\Omega$), amplification factor 1 and time constant 0. (Check the appendix for a detailed figure of the Amplifier and principle behind the measurement of charge Q .)

The surface area A of the capacitor plates is determined by means of their radius and found to be $A = 0.0531 \text{ m}^2$. The distance between the plate capacitor d is set by turning the Vernier scale. (Ask a demonstrator if you do not know how to read a Vernier scale).

1.3.1 Find the linear relation between charge Q and potential V and calculate ϵ_0 .

For this purpose, connect the set-up to the high voltage supply according to Fig.2(a). Ensure that the voltage value is 0 in the display. Set the distance between the two plates $d = 2 \text{ mm}$ using the Vernier scale. You will vary the supply voltage from 0.5kV to 3.5kV in step of 0.5kV. For taking reading in the voltmeter for each value of the supply voltage repeat the following three steps and record your reading in a table (as shown in Table 1).

Step 1: Charge the plate capacitor with the required value of the high voltage supply (e.g. 0.5kV) by turning the knob.

Step 2: Disconnect the high voltage supply (red) cable from the plate. Keep the ground connection unchanged. Insert the amplifier cable(black) to the connector on the plate (to which the high voltage cable was connected) and measure the voltage reading on the voltmeter.

Step 3: After recording your reading, press the discharge button on the amplifier.

Repeat these 3 steps for each value of the supply voltage. Once you finish taking readings, turn the knob to make the voltage supply 0.

Calculate Q according to equation (7) and ϵ_0 according to (8).

Table 1: $A = 0.0531 \text{ m}^2$, $d = 2 \text{ mm}$, $C = 220 \text{ nF}$

Supply voltage V_0 [kV]	0.5	1	1.5	2	2.5	3	3.5
Voltmeter reading V_I [V]							
Q [nC] = CV_I							
ϵ_0 [F/m] = $\frac{d}{A} \frac{Q}{V_0}$							

Question 1

Plot in a mm-graph paper charge Q versus supply voltage V_0 .

Question 2

Explain why the measured value of ϵ_0 is not constant as the expected theoretical value.

Question 3

Using the theoretical value of ϵ_0 , surface area of the plates A , and separation distance d used in the measurement above, calculate the value of the plate capacitor C .

1.3.2 Find the relation between Q and $1/d$

For this purpose, connect the set-up to the high voltage supply according to Fig.2(a). Ensure that the voltage value is 0 in the display. You will vary the separation distance d from 1mm to 3.5mm in step of 0.5mm using the Vernier scale. For taking reading in the

voltmeter for each value of d repeat the following steps and record your reading in a table (as shown in Table 2).

Step 1: Set the separation distance d to the required value (e.g. 1 mm) using Vernier scale.

Step 2: Charge the plate capacitor from the high voltage supply by setting the value at 1.5kV using the turning knob.

Step 3: Disconnect the high voltage supply (red) cable from the plate. Keep the ground connection unchanged. Insert the amplifier cable(black) to the connector on the plate (to which the high voltage cable was connected) and measure the voltage reading on the voltmeter.

Step 4: After recording your reading, press the discharge button on the amplifier and turn the knob on the high voltage supply to zero.

Repeat these steps for each value of d . Once you finish taking readings, turn the knob to make the voltage supply 0.

Table 2: $A = 0.0531\text{m}^2$, $V_0 = 1.5\text{kV}$, $C = 220\text{nF}$

Separation distance d [mm]	1	1.5	2	2.5	3	3.5
Voltmeter reading V_I [V]						
Q [nC] = CV_I						
ϵ_0 [F/m] = $\frac{d}{A} \frac{Q}{V_0}$						
Capacitance of the plate capacitor C_0 [pF]						

Question 4

Plot in a mm-graph paper charge Q versus inverse of the separation distance $1/d$.

Question 5

Calculate the capacitance of the plate capacitor using the theoretical value of ϵ_0 for each value of d .

Separation distance d [mm]	1	1.5	2	2.5	3	3.5
Capacitance of the plate capacitor C_0 [pF]						

Compare these values with those obtained from your measurements in Table 2. Are they same? Comment.

Question 6

If you must find the value of C_0 very accurately, can you use ‘method of image’ or Laplace’s equation? Attempt at home to use one of these two methods to calculate C_0 and compare with your measurement.

1.3.3 Find the relative permittivity ϵ_r of plastic.

For this purpose, connect the set-up to the high voltage supply according to Fig.2(a). Ensure that the voltage value is 0 in the display. You will take two sets of readings –

(i) with the plastic plate thickness of which is $d = 9.8\text{mm}$, (ii) without plastic plate by setting $d = 9.8\text{mm}$.

For both cases, you will vary the supply voltage from 0.5kV to 4kV in step of 0.5kV. For taking reading in the voltmeter for each value of the supply voltage repeat the same three steps described in 1.3.1 and record your reading in a table (as shown in Table 3).

Ensure that when you insert the plastic plate between the two plates, the Vernier scale reading is $d = 9.8$ mm.

Table 3: $A = 0.0531\text{m}^2$, $d = 9.8$ mm, $C = 220\text{nF}$

Supply voltage V_0 [kV]	0.5	1	1.5	2	2.5	3	3.5	4
Voltmeter reading V_I [V]								
Q [nC] = CV_I								
Voltmeter reading V_{I_vac} [V]								
Q_{vac} [nC] = CV_{I_vac}								
$\epsilon_r = \frac{Q}{Q_{vac}}$								
*Capacitance of the plate capacitor with plastic plate C_0 [pF] = $\epsilon_0 \epsilon_r \frac{A}{d}$								
*Capacitance of the plate capacitor without plastic plate C_0 [pF] = $\epsilon_0 \frac{A}{d}$								

*Use the theoretical value of ϵ_0 .

Question 7

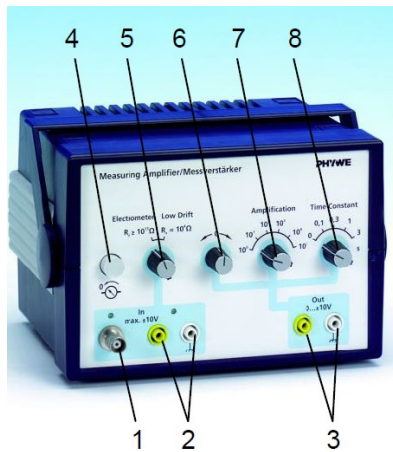
Plot in a mm-graph paper charge Q and Q_{vac} versus supply voltage V_0 .

Question 8

Compare the value of capacitance of the plate capacitor with and without plastic plate and comment.

APPENDIX

Measurement Amplifier



- 1: Input BNC Jack for 'Electrometer' mode. The corresponding LED lights up when switch 5 is set to 'Electrometer'.
2. Jacks for low drift mode, not in use for this experiment.
3. Output jack to connect the voltmeter.
4. Discharge key
5. Operating mode selector
6. zero adjustment potentiometer for zero input voltage.
7. Amplification factor selector
8. Low pass filter time-constant selector.

The principle behind the measurement of charge Q via the amplifier is as follows:

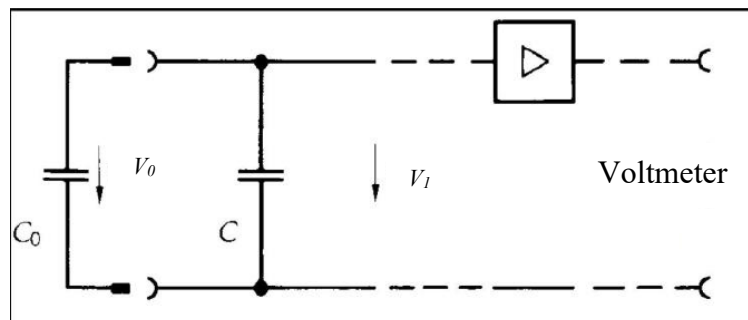


Figure 4: Measurement of charge Q according to step 2.

The charge Q to be measured is on the unknown plate capacitor C_0 and equals to $Q = C_0 V_0$. The discharge button of the amplifier completely discharges the input capacitor C (220nF) and hence, there is no initial charge on the capacitor C . Once, the plate capacitor C_0 is connected to the amplifier (as in step 2 described below), the charge Q is distributed over the both capacitors $Q = (C_0 + C)V_1$. However, $C_0 \ll C$ and hence, it is assumed that $Q = CV_1$. Since, the amplification factor is set as 1, hence, the voltmeter reading is same as V_1 . Thus, using the voltmeter reading and $C = 220\text{nF}$, Q can be calculated.